

Week 10: Multilevel regression and post-stratification

29/03/2026

1 Overview

In this lab we will look at an example of doing multilevel regression with the fertility intentions data. We will also go through how to do post-stratification and MRP.

2 Packages and data

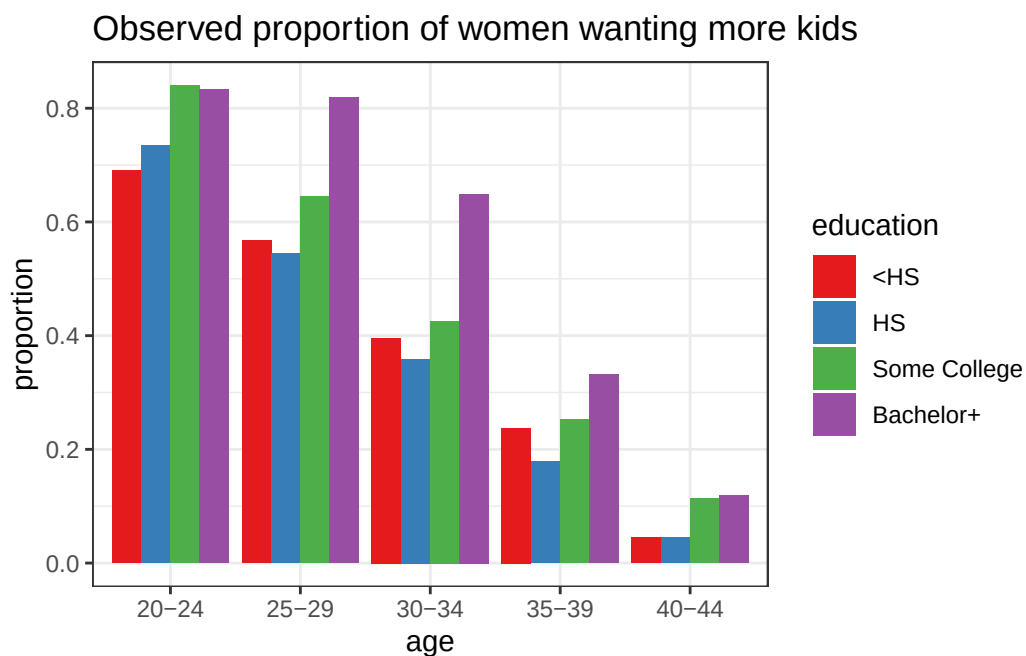
Read in the data from [NSFG](#):

```
library(tidyverse)
library(lme4)
library(rstan)
d <- read_csv("/Users/liangqishen/Desktop/data/intentions.csv")
d <- d |>
  mutate(educ_cat = ifelse(educ == "Bachelor"|educ == "Post-Grad", "Bachelor+", educ)) |>
  mutate(educ_cat = fct_relevel(educ_cat, "<HS", "HS", "Some College", "Bachelor+"))
```

3 Descriptives

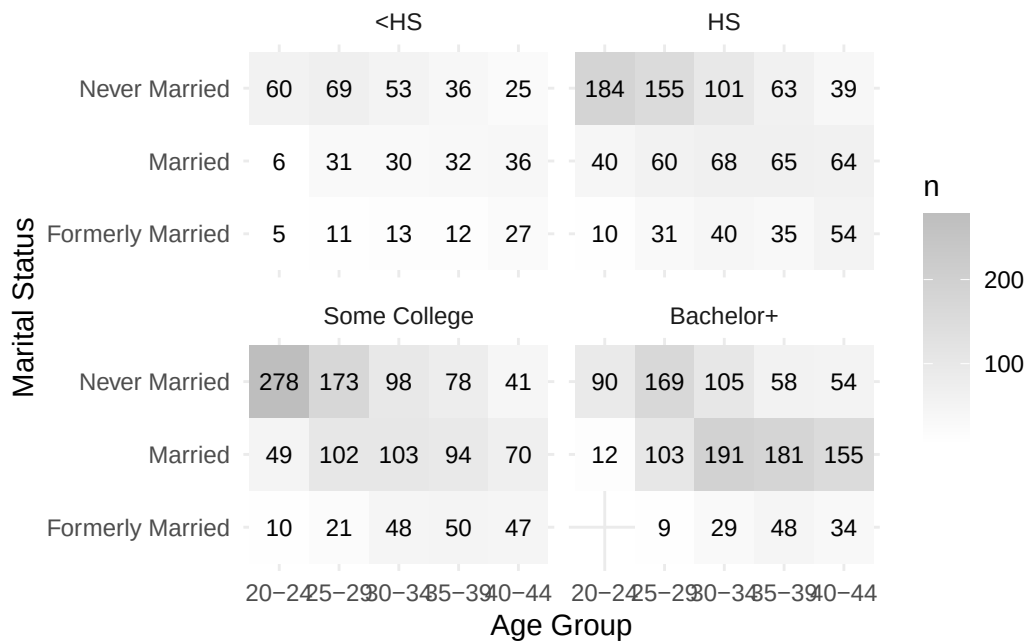
Let's do some brief descriptive plots. Here's the observed proportion of women wanting more kids by education and age:

```
d |>
  group_by(age_gp, educ_cat, wants_more_kids) |>
  tally() |>
  group_by(age_gp, educ_cat) |>
  summarize(prop_wants_more = n[wants_more_kids==1]/sum(n), n = sum(n)) |>
  filter(age_gp!="15-19") |>
  ggplot(aes(age_gp, prop_wants_more, fill = educ_cat)) +
  labs(title = "Observed proportion of women wanting more kids", x = "age", y = "proportion") +
  theme_bw()+
  scale_fill_brewer(palette = "Set1", name = "education")+
  geom_bar(stat = "identity", position = "dodge")
```



Here's a tile plot of the sample sizes:

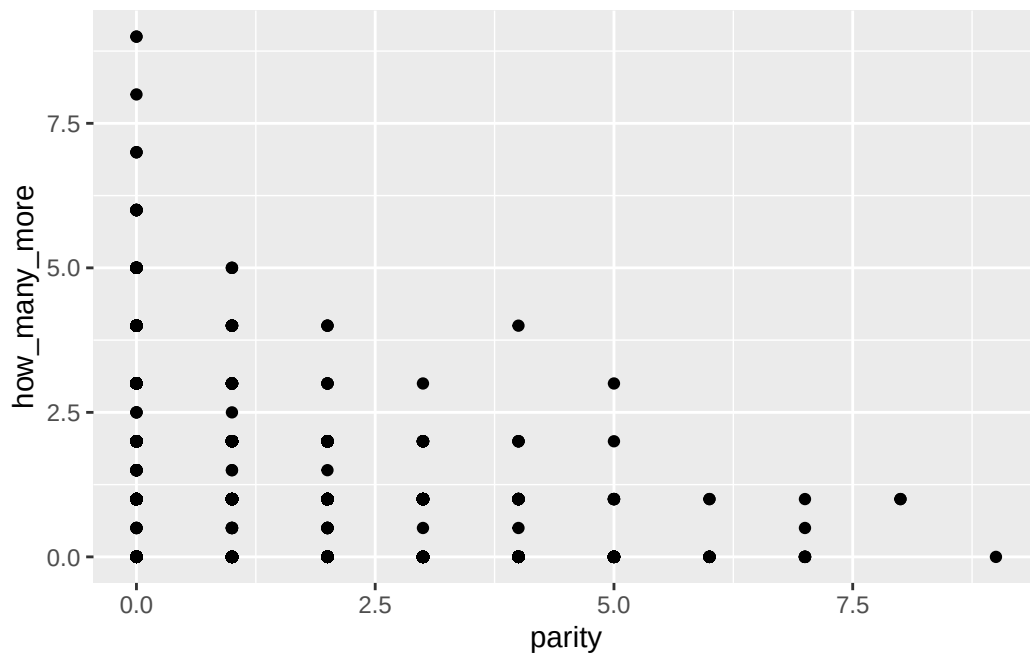
```
d |> group_by(age_gp, marital_formal, educ_cat) |>
  summarize(n = n()) |>
  filter(age_gp!="15-19") |>
  ggplot(aes(x=age_gp, y=marital_formal, fill=n)) +
  geom_tile() +
  scale_fill_gradient(low="white", high="grey") +
  geom_text(aes(label=n), size=3) +
  xlab("Age Group") + ylab("Marital Status") + theme_minimal()+facet_wrap(~educ_cat)
```



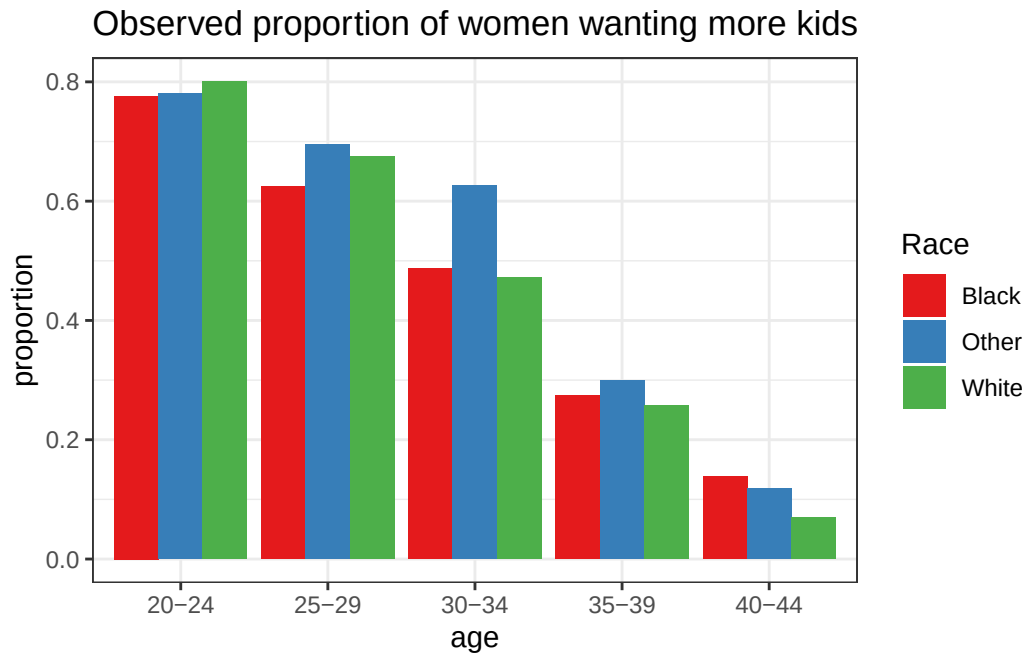
3.1 Question

Do another interesting descriptive chart of the data.

```
d |>
  filter(intention < 10) |>
  mutate(how_many_more = intention - parity) |>
  ggplot(aes(parity, how_many_more)) +
  geom_point()
```



```
d |>
  group_by(age_gp, race_only, wants_more_kids) |>
  tally() |>
  group_by(age_gp, race_only) |>
  summarize(prop_wants_more = n[wants_more_kids==1]/sum(n), n = sum(n)) |>
  filter(age_gp!="15-19") |>
  ggplot(aes(age_gp, prop_wants_more, fill = race_only)) +
  labs(title = "Observed proportion of women wanting more kids", x = "age", y = "proportion") +
  theme_bw()+
  scale_fill_brewer(palette = "Set1", name = "Race")+
  geom_bar(stat = "identity", position = "Dodge")
```



4 Multilevel model

Let's fit the same multilevel model as seen in the lecture using the lme4 package

```
resp4 <- d |> filter(age_gp!="15-19") |> drop_na() |> mutate(age_gp = as_factor(age_gp)) |>
  mutate(age_gp = fct_relevel(age_gp, c("20-24", "25-29", "30-34", "35-39", "40-44")))

mod <- glmer(wants_more_kids ~ marital_formal + (1|age_gp) + (1|educ_cat), data = resp4, fam.
summary(mod)
```

Generalized linear mixed model fit by maximum likelihood (Laplace

Approximation) [glmerMod]

Family: binomial (logit)

Formula: wants_more_kids ~ marital_formal + (1 | age_gp) + (1 | educ_cat)

Data: resp4

AIC	BIC	logLik	-2*log(L)	df.resid
4331.2	4362.6	-2160.6	4321.2	3950

Scaled residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-2.9035 -0.7395 -0.2314 0.6866 5.0382

Random effects:

Groups	Name	Variance	Std.Dev.
age_gp	(Intercept)	1.7471	1.3218
educ_cat	(Intercept)	0.2297	0.4793

Number of obs: 3955, groups: age_gp, 5; educ_cat, 4

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.7963	0.6473	-1.230	0.2186
marital_formalMarried	0.3071	0.1260	2.437	0.0148 *
marital_formalNever Married	0.6298	0.1247	5.051	4.39e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	mrtl_M
mrtl_frmlMr	-0.147	
mrtl_frmlNM	-0.153	0.771

Note! This can also be done in Stan, in a Bayesian framework. This is useful if you want to e.g. play around with the structure of the hierarchical model, which is much easier to control in this framework.

```
stan_data1 <- list(N = nrow(resp4),
  A = nlevels(resp4$age_gp),
  E = nlevels(resp4$educ_cat),
  age = as.numeric(resp4$age_gp),
  edu = as.numeric(resp4$educ_cat),
  x1 = as.numeric(resp4$marital_formal == 'Married'),
  x2 = as.numeric(resp4$marital_formal == 'Never Married'),
  y = resp4$wants_more_kids)

mod1 <- stan(data = stan_data1,
  file = "/Users/liangqishen/Desktop/data/intentions.stan",
  iter = 1000,
  seed = 243,
  refresh=0)
```

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: 'Apple clang version 14.0.0 (clang-1400.0.29.202)'

```

using SDK: 'MacOSX13.1.sdk'
clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/S
In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/R
In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/R
/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
    ~~~~~
1 error generated.
make: *** [foo.o] Error 1

```

```
summary(mod1)$summary[c("beta0", "beta1", "beta2", "sigma_a", "sigma_e"),
                      c("mean", "se_mean", "n_eff", "Rhat")]
```

	mean	se_mean	n_eff	Rhat
beta0	-0.5306225	0.026453829	593.9628	0.9991958
beta1	0.2898085	0.003554718	1201.8785	1.0010173
beta2	0.6139747	0.003413274	1250.6026	1.0005176
sigma_a	1.4061724	0.012152228	962.5477	0.9987282
sigma_e	0.6711102	0.010731276	888.1752	1.0022277

To pull out the estimates from lme4:

```
coef(mod)
```

```

$age_gp
      (Intercept) marital_formalMarried marital_formalNever Married
20-24    0.8455577             0.3070844             0.6297794
25-29    0.1509033             0.3070844             0.6297794
30-34   -0.5704127             0.3070844             0.6297794
35-39   -1.5671327             0.3070844             0.6297794
40-44   -2.8334241             0.3070844             0.6297794

$educ_cat
      (Intercept) marital_formalMarried marital_formalNever Married
<HS             -1.1165369             0.3070844             0.6297794
HS              -1.1969670             0.3070844             0.6297794
Some College    -0.7309838             0.3070844             0.6297794
Bachelor+      -0.1398147             0.3070844             0.6297794

```

```
attr(,"class")
[1] "coef.mer"
```

```
fixef(mod)
```

```
              (Intercept)      marital_formalMarried
              -0.7963065              0.3070844
marital_formalNever Married
              0.6297794
```

```
ranef(mod)
```

```
$age_gp
      (Intercept)
20-24    1.6418643
25-29    0.9472098
30-34    0.2258938
35-39   -0.7708262
40-44   -2.0371176
```

```
$educ_cat
      (Intercept)
<HS          -0.32023040
HS           -0.40066048
Some College  0.06532273
Bachelor+     0.65649179
```

```
with conditional variances for "age_gp" "educ_cat"
```

4.1 Question

Compare the multilevel model with just a standard logistic regression. Which model has a higher AIC? //AIC: sample fit, bigger is better 4295 Generalized linear mixed model is better, has greater AIC

```
resp4 <- d |> filter(age_gp!="15-19") |> drop_na() |> mutate(age_gp = as_factor(age_gp)) |>
  mutate(age_gp = fct_relevel(age_gp, c("20-24", "25-29", "30-34", "35-39", "40-44")))

mod_logistic <- glm(wants_more_kids ~ marital_formal + age_gp + educ_cat, data = resp4, fami
summary(mod_logistic)
```


Call:

```
glm(formula = wants_more_kids ~ marital_formal + age_gp + educ_cat,  
     family = "binomial", data = resp4)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.52573	0.18395	2.858	0.00426	**
marital_formalMarried	0.30102	0.12621	2.385	0.01708	*
marital_formalNever Married	0.62391	0.12486	4.997	5.83e-07	***
age_gp25-29	-0.70322	0.11748	-5.986	2.16e-09	***
age_gp30-34	-1.43055	0.12064	-11.858	< 2e-16	***
age_gp35-39	-2.43411	0.13292	-18.313	< 2e-16	***
age_gp40-44	-3.72135	0.17482	-21.287	< 2e-16	***
educ_catHS	-0.07422	0.13481	-0.551	0.58194	
educ_catSome College	0.40381	0.13078	3.088	0.00202	**
educ_catBachelor+	1.01248	0.13373	7.571	3.71e-14	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 5475.7 on 3954 degrees of freedom
Residual deviance: 4275.0 on 3945 degrees of freedom
AIC: 4295

Number of Fisher Scoring iterations: 5

5 Post stratification

We are interested in obtaining estimates for the proportion of women who want more kids by age. ## Raw proportions by age

Before we do any post-stratification, we can just calculate the raw survey proportions by age:

```
raw_age <- resp4 |>  
  mutate(edu = ifelse(edu == "Bachelor"|edu=="Post-Grad", "Bachelor+", as.character(edu))) |>  
  group_by(age_gp) |>  
  summarize(proportion = mean(wants_more_kids, na.rm = TRUE)) |>  
  mutate(estimate = "raw")
```

5.1 Post-stratification dataset

We are going to use data from the American Community Survey (ACS) as the post-stratification dataset. Note that I got these data from IPUMS USA. Load in ACS data:

```
cell_counts <- read_csv("/Users/liangqishen/Desktop/data/acs_counts.csv")
```

5.2 Question

Compare proportions by age/education/marital group in ACS versus survey.

5.3 Standard post-stratification

Calculate the proportion by group (age, education, marital status) and then re-weight based on the ACS data:

```
ps_age <- resp4 |>
  mutate(edu = ifelse(edu == "Bachelor" | edu == "Post-Grad", "Bachelor+", as.character(edu))) |>
  group_by(age_gp, edu, marital) |>
  summarize(prop = mean(wants_more_kids, na.rm = TRUE)) |>
  filter(marital != "Cohabiting") |>
  left_join(cell_counts) |>
  mutate(prod = prop*n) |>
  group_by(age_gp) |>
  summarize(proportion = sum(prod)/sum(n)) |>
  mutate(estimate = "poststrat")
```

5.4 MRP

Get predicted proportions for each group:

```
age_groups <- rownames(coef(mod)$age_gp)
educ_groups <- rownames(coef(mod)$educ_cat)
marital_groups <- c("Formerly Married", "Married", "Never Married")

pred_data <- expand_grid(age_gp = age_groups, educ_cat = educ_groups, marital_formal = marital_groups)

pred_mrp <- plogis(predict(mod, pred_data))

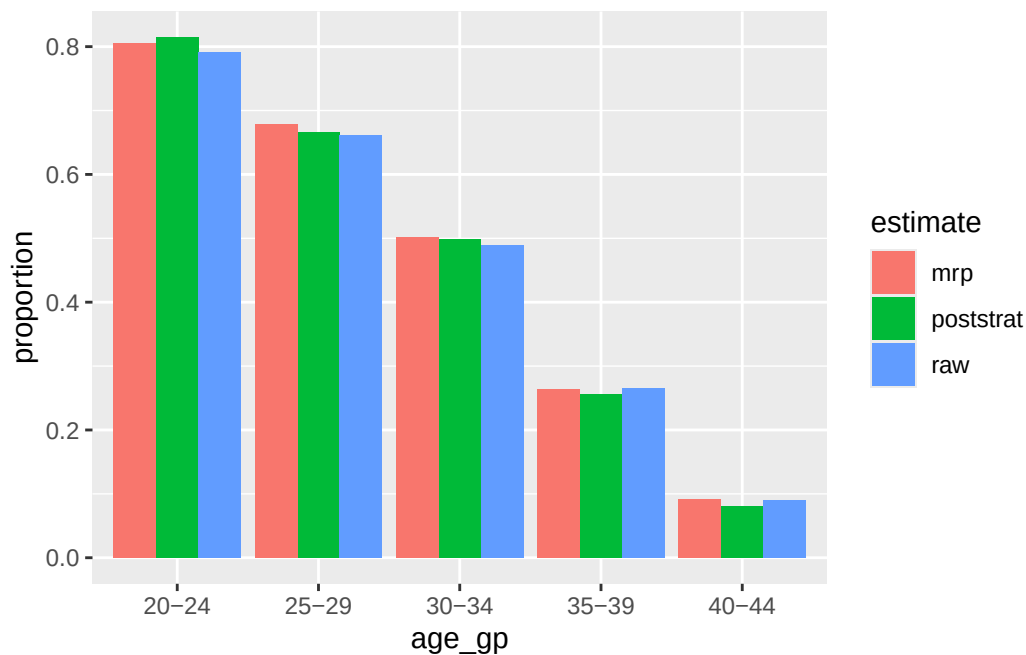
pred_prop <- cbind(pred_data, pred_mrp)
```

Now post-stratify by age:

```
mrp_age <- pred_prop |>
  rename(edu = educ_cat, marital = marital_formal) |>
  left_join(cell_counts) |>
  mutate(prod = pred_mrp*n) |>
  group_by(age_gp) |>
  summarize(proportion = sum(prod)/sum(n)) |>
  mutate(estimate = "mrp")
```

Compare on a plot:

```
ps_age |>
  bind_rows(mrp_age) |>
  bind_rows(raw_age) |>
  ggplot(aes(age_gp, proportion, fill = estimate))+
  geom_bar(stat = "identity", position = "dodge")
```



5.5 Question

Do the same thing but compare across education groups.

5.6 Raw proportions by age

Before we do any post-stratification, we can just calculate the raw survey proportions by age:

```
raw_edu <- resp4 |>
  mutate(edu = ifelse(edu == "Bachelor"|edu=="Post-Grad", "Bachelor+", as.character(edu))) |>
  group_by(edu) |>
  summarize(proportion = mean(wants_more_kids, na.rm = TRUE)) |>
  mutate(estimate = "raw")
```

5.7 Standard post-stratification

Calculate the proportion by group (age, education, marital status) and then re-weight based on the ACS data:

```
ps_edu <- resp4 |>
  mutate(edu = ifelse(edu == "Bachelor"|edu=="Post-Grad", "Bachelor+", as.character(edu))) |>
  group_by(age_gp, edu, marital) |>
  summarize(prop = mean(wants_more_kids, na.rm = TRUE)) |>
  filter(marital != "Cohabiting") |>
  left_join(cell_counts) |>
  mutate(prod = prop*n) |>
  group_by(edu) |>
  summarize(proportion = sum(prod)/sum(n)) |>
  mutate(estimate = "poststrat")
```

5.8 MRP

Get predicted proportions for each group:

```
edu_groups <- rownames(coef(mod)$edu)
educ_groups <- rownames(coef(mod)$educ_cat)
marital_groups <- c("Formerly Married", "Married", "Never Married")

pred_data <- expand_grid(age_gp = age_groups, educ_cat = educ_groups, marital_formal = marital_groups)

pred_mrp <- plogis(predict(mod, pred_data))

pred_prop <- cbind(pred_data, pred_mrp)
```

Now post-stratify by age:

```
mrp_edu <- pred_prop |>
  rename(edu = educ_cat, marital = marital_formal) |>
  left_join(cell_counts) |>
  mutate(prod = pred_mrp*n) |>
  group_by(edu) |>
  summarize(proportion = sum(prod)/sum(n)) |>
  mutate(estimate = "mrp")
```

Compare on a plot:

```
ps_edu |>
  bind_rows(mrp_edu) |>
  bind_rows(raw_edu) |>
  ggplot(aes(edu, proportion, fill = estimate))+
  geom_bar(stat = "identity", position = "dodge")
```

